

AgriSmart AI – Literature Review

Project: AgriSmart AI – Intelligent Crop Recommendation System **Document Type:** Literature Review (DEPI Project Documentation) **Focus:** Machine Learning for Crop Recommendation

1. Introduction

Agriculture is the foundation of global food production, and the choice of which crop to cultivate strongly affects productivity, resource use, and farm income. Selecting a suitable crop is difficult because it depends on several interacting factors, including soil nutrients, soil acidity, temperature, humidity, and rainfall. Farmers have traditionally relied on experience and local practice, which may not capture how these factors combine under specific field conditions. Kulkarni et al. note that a key setback to productivity is that farmers do not always choose the right crop for cultivation [2].

Soil and environmental parameters provide measurable inputs for data-driven crop selection. Machine Learning (ML) can learn the relationship between these inputs and the crops that suit them, and can then recommend a crop for new conditions. This review examines previous ML-based crop recommendation studies, the parameters they use, and the role of Explainable AI (XAI), and it then positions the AgriSmart AI project in relation to this work.

2. Machine Learning for Crop Recommendation

2.1 Suitability of ML Models for Tabular Agricultural Data

Crop recommendation datasets are typically tabular: each record contains a small number of numeric soil and climate measurements and a categorical crop label. For this type of data, tree-based models are widely used. Grinsztajn et al. benchmarked tree-based models and deep learning models on typical tabular datasets and found that tree-based models remained state of the art on medium-sized data, which are the dataset sizes common in crop recommendation [8]. This general finding supports the use of Random Forest and boosting methods, such as XGBoost, as strong candidates for the AgriSmart AI task. However, [8] is not a crop-specific study, so it indicates suitability rather than proving performance on any particular crop dataset.

2.2 Ensemble Approaches

A recurring theme in the reviewed crop recommendation literature is the use of ensemble methods that combine several classifiers.

Pudumalar et al. proposed a recommendation system based on an ensemble with majority voting, using Random Tree, CHAID, K-Nearest Neighbor (KNN), and Naive Bayes as base learners, to recommend a crop for site-specific conditions [1]. Kulkarni et al. also used majority voting, with Random Forest, Naive Bayes, and linear Support Vector Machine (SVM) as base learners. Their system classified soil data into recommendable crop types (Kharif and Rabi) and reported an average classification accuracy of 99.91% [2].

Acharya et al. compared individual classifiers with a soft-voting ensemble of Naive Bayes, SVM, Decision Tree, and Random Forest. They reported accuracies of 98.38% for Decision Tree, 98.90% for Random Forest, 98.14% for Naive Bayes, and 98.50% for SVM, with an ensemble accuracy of 98.99%, and they confirmed robustness through cross-validation [3]. The individual models were already close to each other, so the ensemble gain reported there is small in absolute terms.

Ahmed et al. extended this direction to stacking. They proposed feature fusion with a stacking ensemble of 18 classifiers, evaluated on two datasets (one with nearly 28,242 records), and reported accuracies between 98.4% and 99.54% together with reduced overfitting for their proposed models [4].

2.3 Strengths and Limitations

Strengths. The reviewed studies consistently report high accuracy for tree-based and ensemble classifiers, and several show that combining models gives a measurable, if sometimes modest, improvement over single models [2]–[4]. Simple classifiers such as Decision Tree, Naive Bayes, and SVM already perform well on these datasets [3], which suggests that the task is learnable from a small set of tabular features.

Limitations. First, accuracy values near 99% across different studies suggest that the evaluated datasets may be easy to separate, and such results should not be assumed to transfer to real field conditions. Second, ensembles with many base learners increase complexity and make it harder to understand individual predictions [4]. Third, the abstracts of the reviewed accuracy-focused studies [1]–[4] do not report explanations of individual predictions. Finally, the reviewed studies provide limited direct evidence on Logistic Regression, and evidence for gradient boosting or XGBoost in crop recommendation specifically is limited in this review; support for these models comes mainly from general tabular benchmarks [8]. These models are therefore reasonable candidates to compare in AgriSmart AI, not models with proven superiority.

3. Soil & Environmental Parameters

Crop recommendation studies commonly use soil nutrient and climate variables as inputs. Ahmed et al. describe recommendations that take climate, weather, and soil nutrients into account [4], and Bouni et al. describe contributions of soil quality, climatic variables, and historical crop performance to predictions [5]. Kulkarni et al. base recommendations on soil-specific type and characteristics [2].

The parameters most commonly used, and the general reasons they matter for crop suitability, are summarized below.

Parameter	Relevance to Crop Suitability
Nitrogen (N)	Supports leaf and vegetative growth; demand differs between crops
Phosphorus (P)	Supports root development and flowering
Potassium (K)	Supports water regulation and general plant health
pH	Influences nutrient availability and the range of crops a soil can support
Temperature	Determines which crops can grow and complete their cycle
Humidity	Affects crop water balance and disease conditions
Rainfall	Determines natural water supply and irrigation need

The agronomic explanations in this table are general background; the cited studies establish that such variables are used as model inputs, and the models learn their relationship to crop labels from data. No single parameter determines suitability, which is why a multivariate ML approach is appropriate.

4. Explainable AI in Agriculture

High-accuracy models such as ensembles and boosted trees are often treated as black boxes. In crop recommendation, a user who receives only a crop name has no way to judge whether the recommendation reflects sensible agronomic reasoning. Bouni et al. address this by combining Random Forest with SHAP and LIME to provide transparent explanations for each crop recommendation, noting that this fosters user trust, particularly when predictions diverge from established expert opinion [5].

Three explanation approaches are relevant to this project:

- **Feature importance.** Tree-based models such as Random Forest can rank features by how much they contribute overall. This gives a global view of which parameters matter most but does not explain a single prediction.
- **SHAP.** Lundberg and Lee proposed SHAP (SHapley Additive exPlanations), a unified framework that assigns each feature an importance value for a particular prediction, based on game-theoretic Shapley values [6]. SHAP supports both local explanations (why this crop for this input) and global summaries across many predictions.
- **LIME.** Ribeiro et al. proposed LIME (Local Interpretable Model-agnostic Explanations), which explains an individual prediction of any classifier by approximating the model locally with an interpretable model [7].

Interpretability is useful in crop recommendation for three reasons. It allows users to check whether the influential factors, such as nutrient levels or rainfall, are plausible for the recommended crop. It helps developers detect problems such as a model relying on an irrelevant feature. It also supports trust and adoption by non-technical users. A SHAP explanation, for example, can show that a high rainfall value and a particular nitrogen level pushed the model toward one crop and away from another, allowing the user to understand the recommendation instead of accepting it blindly.

5. Comparison of Previous Studies

The table below summarizes the studies reviewed in Sections 2 and 4 using only information reported in their abstracts. "Not specified" means the information was not stated in the abstract and is not assumed here.

Study	Dataset	Features	Models	Main Result	Contribution
Pudumalar et al. [1]	Not specified	Soil and site-specific parameters	Random Tree, CHAID, KNN, Naive Bayes; majority voting	Reported high accuracy and efficiency (no figure cited here)	Ensemble voting for site-specific crop recommendation
Kulkarni et al. [2]	Soil dataset (Kharif and Rabi crop types)	Soil type and characteristics	Random Forest, Naive Bayes, linear SVM; majority voting	Average accuracy 99.91%	Ensemble voting for crop-type recommendation

Study	Dataset	Features	Models	Main Result	Contribution
Acharya et al. [3]	Not specified	Not specified	Naive Bayes, SVM, Decision Tree, Random Forest; soft voting	Ensemble 98.99% vs. 98.14–98.90% for individual models	Comparative study with cross-validation
Ahmed et al. [4]	Two datasets (one with about 28,242 records)	Climate, weather, and soil nutrients	Stacking ensemble of 18 classifiers with feature fusion	Accuracy 98.4%–99.54%; reduced overfitting reported	Feature fusion and stacking for crop recommendation
Bouni et al. [5]	Agricultural parameters, historical yields, and environmental conditions	Soil quality, climatic variables, historical crop performance	Random Forest with SHAP and LIME	Reported improved accuracy over traditional methods (no figure in abstract)	Global and local explanations for crop recommendations

The reported results come from different datasets and evaluation settings and are therefore not directly comparable.

6. Research Gap

The reviewed literature shows that ML, particularly ensemble and tree-based methods, can achieve high accuracy on crop recommendation data [1]–[4]. Based on this review, the following gaps are identified, stated cautiously because the review covers a limited number of studies:

- 1. Emphasis on predictive accuracy.** Most reviewed studies [1]–[4] concentrate on classification performance and ensemble design, and their abstracts do not report explanation of individual predictions.
- 2. Explanation of individual predictions.** Studies that do add XAI, such as [5], show that SHAP and LIME can be applied to crop recommendation, but explanation remains a less common component than accuracy reporting in this review.
- 3. Integration into a simple interface.** Among the reviewed abstracts, the connection between prediction, explanation, and a simple user-facing application is not a prominent theme. This is an observation about the reviewed abstracts and not a claim about all published work.

4. Benchmark versus real-world use. Very high accuracies on benchmark-style datasets do not guarantee equivalent performance under real agricultural conditions, where soil and climate measurements vary and may be noisy. This is a general caution that applies to the field.

These gaps are modest. They indicate where a project can add practical value, not that existing work is deficient.

7. AgriSmart AI Positioning

AgriSmart AI is a student project that builds on established methods rather than proposing a new algorithm. Based on the reviewed literature, it is positioned as follows:

Aspect	AgriSmart AI Plan	Relation to Reviewed Literature
Task	Crop recommendation from soil and environmental parameters	Same task as [1]–[5]
Models	Compare multiple ML models, including tree-based and boosting methods	Consistent with the model families studied in [1]–[4] and [8]
Evaluation	Systematic evaluation and hyperparameter tuning	Standard practice; cross-validation is used in [3]
Explainability	SHAP-based explanation of predictions	Builds on SHAP [6] and on its use in crop recommendation [5]
Delivery	A simple user-facing application	Addresses the integration gap in Section 6 as a practical objective

AgriSmart AI does not claim scientific novelty or superior accuracy compared with the reviewed studies. Any statement about its performance will depend on experiments conducted during the project, and any comparison with prior work will be made only if the datasets and evaluation conditions are comparable. Its contribution is expected to be practical: an end-to-end, documented, and explainable implementation of established techniques.

8. Conclusion

The reviewed studies show that ML is well suited to crop recommendation from tabular soil and environmental data. Tree-based and ensemble models report consistently high accuracy [1]–[4], and general tabular-data evidence supports tree-based models as

strong candidates [8]. At the same time, high accuracy alone does not tell a user why a crop is recommended. XAI methods such as SHAP [6] and LIME [7], applied to crop recommendation in [5], address this by showing which inputs drive a prediction.

These findings support the choice of ML for AgriSmart AI, because the data are structured and the task is a well-studied classification problem, and the choice of XAI, because interpretable recommendations are more useful and trustworthy for end users. The review also confirms that results from other studies cannot be assumed for this project, so model performance will be established through the project's own evaluation.

9. References

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